AS + A2 · Units 3 & 6 · WCH13 / WCH16 | Practical Skills

# Core practicals and skills

Units 3 and 6 test practical chemistry on paper: you may be asked to describe a method, explain why each step is done, process results, or suggest improvements. This page summarises every core practical and the skills shared between them. The worked chemistry for each one is in its own topic page.

## Skills used in every practical

### Safety

- Name the hazard and give a precaution that matches it: eye protection for irritants and corrosives (acids, alkalis); gloves and a fume cupboard for toxic substances or vapours; no naked flames near flammable liquids (heat with an electric heater or water bath); keep oxidising agents away from flammable materials.
- A hazardous substance at low concentration or in small amounts carries a lower risk than the pure substance.

### Measuring accurately

- **Mass by difference**: weigh the weighing bottle with the solid, tip the solid out, then reweigh the empty bottle. The difference is exactly the mass transferred.
- **Volumes**: a volumetric pipette or burette is more precise than a measuring cylinder. Read the bottom of the meniscus at eye level (the top for dark solutions such as $KMnO_4$), and record burette readings to the nearest 0.05 $cm^3$.
- **Uncertainty**: a reading is ± half the smallest division (or the value printed on the apparatus); a measurement made from two readings, such as a titre or a temperature change, has twice that. Percentage uncertainty = uncertainty ÷ value × 100; add the percentages for every piece of apparatus to estimate the total. Making the measured quantity larger reduces the percentage uncertainty.
- **Repeat** measurements and discard anomalies; in titrations, keep going until you have **concordant** titres within 0.10 $cm^3$ and average only those.

### Graphs

- Plot the independent variable on the x-axis, use most of the grid, label axes with quantity and unit, and draw a line or smooth curve of best fit.
- Find a gradient from a large triangle using points on the line, not data points. A tangent to a curve gives the rate at that moment; the tangent at t = 0 gives the initial rate.
- Extrapolate a cooling curve back to the time of mixing to correct for heat loss in calorimetry.

## Drawing apparatus

Draw a continuous outline with no lines across joints, and make sure the apparatus would work: a condenser has a sealed outer jacket with water in at the bottom and out at the top; a heated system must never be sealed; a gas collection system must not have gaps where gas could escape.

# AS core practicals

## CP1 · Molar volume of a gas

React known masses of calcium carbonate (about 0.05 to 0.30 g) with excess ethanoic acid in a flask connected to a gas syringe, or collect the gas over water in an upturned measuring cylinder. Plot volume of $CO_2$ against mass of $CaCO_3$; the gradient gives volume per gram, and molar volume = volume of gas ÷ moles of $CaCO_3$.

- **Errors**: gas escapes before the bung is replaced; $CO_2$ is slightly soluble in water; the syringe plunger can stick. Note room temperature and pressure, because gas volume depends on both.
- **Related methods**: finding $M_r$ of a volatile liquid by injecting a known mass into a heated gas syringe and using $pV = nRT$; finding water of crystallisation by heating a hydrated salt in a crucible to constant mass.

## CP2 · Enthalpy change using Hess's law

Find an enthalpy change that cannot be measured directly from two that can, for example the enthalpy change of hydration of copper(II) sulfate from the enthalpy changes of solution of anhydrous and hydrated copper(II) sulfate.

- **Method**: put a measured volume of water in a polystyrene cup (insulation), record the temperature every minute for a few minutes, add the weighed solid at a known minute, stir, and keep recording for about 15 minutes.
- **Analysis**: plot temperature against time, extrapolate both lines to the moment of mixing, and read $\Delta T$. Then $q = mc\Delta T$ ($m$ = mass of solution, $c = 4.18$ J $g^{-1}$ $K^{-1}$), $\Delta H = -q \div$ moles, with a sign and units in kJ $mol^{-1}$.
- **Errors**: heat lost to the surroundings, the heat capacity of the cup ignored, assuming the solution has the density and heat capacity of water, incomplete dissolving. **Enthalpy of combustion** by spirit burner has further errors: incomplete combustion and evaporation of the fuel.

## CP3 and CP4 · Making a standard solution and titrating

- **Standard solution**: weigh the solid by difference, dissolve it in distilled water in a beaker, pour into a volumetric flask through a funnel, rinse the beaker, rod and funnel into the flask, make up to the mark with a dropping pipette (bottom of the meniscus on the line), then stopper and invert several times.
- **Titration**: rinse the pipette with the solution it will deliver and the burette with its solution; rinse the conical flask with distilled water only. Fill the burette jet; remove the funnel before titrating. Add a few drops of indicator, swirl, add dropwise near the end point over a white tile, and repeat for concordant results.
- **Indicators**: phenolphthalein (colourless in acid, pink in alkali) for titrations with a strong base; methyl orange (red in acid, yellow in alkali, orange at the end point) for titrations with a strong acid.

- Adding distilled water to wash the flask's sides does not change the moles present, so it does not affect the titre.

## CP5 · Rates of hydrolysis of halogenoalkanes

Put equal amounts of 1-chlorobutane, 1-bromobutane and 1-iodobutane into tubes, add ethanol (as a common solvent) and aqueous silver nitrate, warm in a water bath at the same temperature, and time how long each precipitate takes to appear. Water hydrolyses the C–X bond, and the halide ion released precipitates as AgX: yellow AgI forms first, then cream AgBr, then white AgCl. The weakest bond (C–I) breaks fastest, so bond enthalpy, not polarity, controls the rate.

## CP6 · Making 2-chloro-2-methylpropane

Shake 2-methylpropan-2-ol with concentrated hydrochloric acid in a separating funnel, releasing pressure often. Separate and discard the aqueous layer; wash the organic layer with sodium hydrogencarbonate solution to remove acid (releasing $CO_2$ pressure), separate again, dry with anhydrous sodium sulfate, and distil, collecting the fraction boiling near 51 °C. Cyclohexene can be made similarly by dehydrating cyclohexanol with concentrated phosphoric acid and distilling it off.

## CP7 · Oxidising propan-1-ol

- **To propanal**: add acidified sodium dichromate(VI) a drop at a time to warm propan-1-ol and distil off the aldehyde as it forms (it has the lowest boiling point), so it is not oxidised further. Use a limited amount of oxidising agent.
- **To propanoic acid**: use an excess of oxidising agent and heat under reflux for about 15 minutes, then distil the acid off.
- Orange dichromate(VI) turns green as $Cr^{3+}$ forms. Dichromate is toxic and carcinogenic, so wear gloves; concentrated sulfuric acid is corrosive.

## CP8 · Analysing inorganic and organic unknowns

- **Flame tests** (nichrome wire cleaned in concentrated HCl): lithium red, sodium yellow, potassium lilac, calcium brick red, strontium red, barium apple green; magnesium gives no colour.
- **Cations with NaOH(aq)**: $Mg^{2+}$ white precipitate; $Ca^{2+}$ slight white precipitate; transition metal hydroxides coloured (see Topic 17). **Ammonium ions**: warm with NaOH(aq); ammonia turns damp red litmus blue.
- **Anions**: carbonate fizzes with acid, giving $CO_2$ that turns limewater milky; sulfate gives a white precipitate with barium chloride in hydrochloric acid (the acid removes carbonate, which would also precipitate); halides give AgX precipitates with silver nitrate in nitric acid, distinguished by their solubility in dilute and concentrated ammonia.
- **Organic tests**: see the functional-group table in Topic 20.

# A2 core practicals

## CP9a · Rate of the iodine–propanone reaction by titration

$CH_3COCH_3 + I_2 \rightarrow CH_3COCH_2I + H^+ + I^-$, catalysed by acid. Mix propanone and sulfuric acid, add iodine solution and start the clock. Every few minutes pipette out a sample and **quench** it in sodium hydrogencarbonate solution, which neutralises the acid catalyst and stops the reaction, then titrate the remaining iodine with sodium thiosulfate using starch near the end point.

A graph of $[I_2]$ against time is a straight line: the rate does not change as iodine is used up, so the reaction is **zero order** in iodine. Propanone and acid are in large excess so their concentrations stay almost constant. The rate equation is rate = $k[CH_3COCH_3][H^+]$, so iodine is not involved in the rate-determining step.

## CP9b · A clock reaction

Hydrogen peroxide oxidises iodide ions to iodine; a small, fixed amount of thiosulfate turns the iodine straight back into iodide. When all the thiosulfate is used up, iodine builds up and turns the starch suddenly blue-black. The time t to reach this fixed point is measured; 1/t is proportional to the initial rate. Repeat with different concentrations of one reactant, keeping the total volume the same with water, and plot log(1/t) against log(concentration): the gradient is the order with respect to that reactant.

## CP10 · Activation energy

Time a reaction that reaches a fixed end point, such as bromide and bromate(V) ions in acid producing bromine that bleaches methyl red after it has reacted with a fixed amount of phenol, at several temperatures between about 20 and 60 °C, warming the solutions separately in a water bath before mixing. The rate constant is proportional to 1/t, so the Arrhenius equation gives

$$\ln \frac{1}{t} = -\frac{E_a}{R} \cdot \frac{1}{T} + \text{constant}$$

Plot ln(1/t) against 1/T (in $K^{-1}$): the gradient is $-E_a/R$, so $E_a$ = −gradient × 8.31 J $K^{-1}$ $mol^{-1}$.

## CP11 · $K_a$ of a weak acid

Calibrate a pH meter with buffer solutions (it drifts on storage). Either record a full pH curve as alkali is added, or titrate to find the neutralisation volume, then add exactly half that volume to a fresh sample of the acid: at half-neutralisation pH = $pK_a$, so $K_a = 10^{-pH}$. Keep the temperature constant, as $K_a$ depends on it (see Topic 14).

## CP12 · Electrochemical cells

Clean metal strips with emery paper, dip each into 1.00 mol $dm^{-3}$ solution of its ions, join the beakers with a salt bridge of filter paper soaked in saturated potassium nitrate, and connect the metals through a high-resistance

voltmeter. Use a platinum electrode where there is no metal in the half-equation, for example a mixture of $Fe^{2+}$ and $Fe^{3+}$ both at 1.00 mol $dm^{-3}$ (see Topic 16).

## CP13a and CP13b · Redox titrations

- **Iron(II) with manganate(VII)**: grind iron tablets, dissolve in dilute sulfuric acid, make up to a known volume, and titrate portions with potassium manganate(VII) to the first permanent pink. No indicator is needed. Use dilute sulfuric acid, not HCl or $HNO_3$.
- **Iodine with thiosulfate**: liberate iodine from an oxidising agent with excess KI, titrate with sodium thiosulfate until pale yellow, then add starch and continue until the blue-black colour disappears.
- Full methods, equations and worked calculations are in Topic 16.

## CP14 · Making a transition metal complex

Dissolve weighed hydrated copper(II) sulfate in a little hot water, then in a fume cupboard add concentrated ammonia until the deep blue solution forms. Add ethanol, in which the complex is much less soluble, and cool in ice to crystallise tetraamminecopper(II) sulfate: $CuSO_4 \cdot 5H_2O + 4NH_3 \rightarrow [Cu(NH_3)_4(H_2O)]SO_4 + 4H_2O$. Filter with a Büchner funnel, wash with a little cold ethanol, dry between filter papers and weigh to find the percentage yield. Yield is lost in filtering and washing, and because some product stays in solution.

## CP15 · Further analysis of unknowns

Combines the tests of CP8 with the A2 chemistry: reactions of transition metal ions with NaOH(aq) and $NH_3$(aq), tests for aldehydes and ketones (2,4-DNPH, Tollens', Fehling's), the iodoform test, and tests for carboxylic acids, phenols and acyl chlorides. Interpreting spectra (IR, mass spectra and NMR) may be combined with these.

## CP16 · Preparing aspirin

Warm 2-hydroxybenzoic acid with ethanoic anhydride and a few drops of phosphoric acid catalyst under reflux; the phenol group is esterified. Ethanoic anhydride is used rather than ethanoyl chloride because it is cheaper, less corrosive, reacts less violently and gives off ethanoic acid rather than HCl. Add cold water to destroy excess anhydride and precipitate the crude aspirin, then **recrystallise** from a minimum of hot ethanol and water: hot filter, cool slowly then in ice, filter under reduced pressure, wash with a little cold water and dry. Check purity by melting point (pure aspirin melts sharply at about 136 °C) and by thin-layer chromatography (a single spot). The same purification is used for methyl 3-nitrobenzoate made by nitrating methyl benzoate below 10 °C.

# Other experiments that often appear

- **Finding $K_c$ for an esterification** by titrating the acid left at equilibrium (see Topic 13).
- **Colorimetry**: make a calibration curve with solutions of known concentration using a filter of the complementary colour, then read off the unknown (see Topic 17).
- **Autocatalysis** of manganate(VII) with ethanedioate, followed by quenching samples in KI and titrating the iodine (see Topic 17).

- **Thin-layer chromatography** and $R_f$ values (see Topic 15); **melting and boiling points** and **purifying liquids** (see Topic 20).
- **Following rates continuously**: gas volume with a syringe, loss of mass as gas escapes, colorimetry, conductivity, or polarimetry for reactions that change optical activity. Taking samples and titrating them is the alternative when no property can be monitored continuously.

## Quick checks

QUICK CHECK

**In CP2 the temperature is recorded for several minutes before and after mixing. Why?**

- to check the thermometer works
- so the lines can be extrapolated to the time of mixing, correcting for heat lost while the reaction happens
- to make sure the solid has dissolved
- to measure the heat capacity of the cup

Heat leaks in or out throughout the experiment. Extrapolating both lines to the moment of mixing estimates the temperature change there would have been with no heat exchange.

QUICK CHECK

**Why are samples in CP9a added to sodium hydrogencarbonate solution?**

- to react with the iodine
- to act as an indicator
- to neutralise the acid catalyst, stopping the reaction before titration
- to dilute the propanone

Quenching freezes the composition at the moment of sampling. Without the acid catalyst the reaction effectively stops, so the titre reflects the iodine present at that time.

QUICK CHECK

**An Arrhenius plot of ln(1/t) against 1/T has a gradient of −6000 K. What is the activation energy?**

- 6.0 kJ $mol^{-1}$
- 49.9 kJ $mol^{-1}$
- −49.9 kJ $mol^{-1}$
- 722 kJ $mol^{-1}$

$E_a$ = −gradient × R = 6000 × 8.31 = 49 900 J $mol^{-1}$ = 49.9 kJ $mol^{-1}$. Activation energies are always positive.

QUICK CHECK

**Why is ethanol added in CP14 when making tetraamminecopper(II) sulfate?**

- it reacts with the ammonia
- the complex is much less soluble in ethanol, so it crystallises out
- it acts as a catalyst
- it oxidises copper(II) to copper(III)

Adding a solvent in which the product is poorly soluble lowers its solubility in the mixture, so more of it crystallises and the yield rises.

QUICK CHECK

**In CP5, which halogenoalkane gives a precipitate fastest, and why?**

- 1-chlorobutane, because the C–Cl bond is most polar
- 1-bromobutane, because AgBr is least soluble
- 1-iodobutane, because the C–I bond is weakest
- all the same, because they have the same chain length

The rate depends mainly on how easily the C–X bond breaks. C–I has the lowest bond enthalpy, even though it is the least polar.

QUICK CHECK

**A titre of 22.50 $cm^3$ is measured with a burette read to ±0.05 $cm^3$. What is the percentage uncertainty in the titre?**

- 0.22%
- 0.44%
- 0.05%
- 2.2%

Two readings make the titre, so its uncertainty is ±0.10 $cm^3$: 0.10 ÷ 22.50 × 100 = 0.44%. 0.22% counts only one reading.